NOTES, QUESTIONS, AND NUMERICAL EXPERIMENTS WITH RANDOM UNITARY MATRICES

N. CHRISTOPHER PHILLIPS

This is a very roughly written collection of information collected that is related to questions such as the following: Given two random large unitary matrices, how close are they likely to be to being unitarily equivalent? Given two random approximately commuting large unitary matrices, how likely are they to be close to exactly commuting unitary matrices?

A suggested general reference for random matrices, which I have not seen, is [10]. I have been warned that the proofs are not always rigorous.

Throughout, unless otherwise specified, matrices will have complex entries. We denote the set of $n \times n$ matrices by M_n. If

$$a = \big(a_{j,k} \big)_{j,k=1}^{n} \in M_n,$$

then its adjoint a^* is, as usual, the conjugate transpose, with (j,k) entry equal to $\overline{a_{k,j}}$. A matrix $a \in M_n$ is selfadjoint if $a^* = a$, unitary if $a^* a = a a^* = 1$, and normal if $a^* a = a a^*$. The group of unitary matrices in M_n is denoted by $U(M_n)$. We recall that normal matrices are unitarily diagonalizable: if $a \in M_n$ is normal, then there is a unitary $u \in M_n$ such that $u a u^*$ (which is the same thing as $u a u^{-1}$) is diagonal. The diagonal entries are, of course, the eigenvalues of a.

We identify M_n with the algebra of linear maps from $\mathbb{C}^n$ to $\mathbb{C}^n$ in the usual way. We write an element $\xi \in \mathbb{C}^n$ in coordinates as $\xi = (\xi_1, \xi_2, \ldots, \xi_n)$ with $\xi_1, \xi_2, \ldots, \xi_n \in \mathbb{C}$, often without explicitly saying that the ξ_k are the coordinates of ξ. On $\mathbb{C}^n$, we use the usual Euclidean norm: if $\xi \in \mathbb{C}^n$, then

$$\|\xi\|_2 = \sqrt{|\xi_1|^2 + |\xi_2|^2 + \cdots + |\xi_n|^2}.$$

The scalar product of $\xi, \eta \in \mathbb{C}^n$ is

$$\langle \xi, \eta \rangle = \xi_1 \overline{\eta_1} + \xi_2 \overline{\eta_2} + \cdots + \xi_n \overline{\eta_n}.$$

We have the usual properties; in particular, for $\xi, \eta \in \mathbb{C}^n$ we have $\|\xi + \eta\|_2 \leq \|\xi\|_2 + \|\eta\|_2$. We sometimes abbreviate $\|\xi\|_2$ to $\|\xi\|$.

We define distances in M_n using the operator norm coming from the Euclidean norm on $\mathbb{C}^n$. That is, for $a \in M_n$,

$$\|a\| = \sup \big\{ \|a\xi\|_2 \colon \xi \in \mathbb{C}^n, \ \|\xi\|_2 \leq 1 \big\}.$$

This expression has the following important properties. (The first three constitute the definition of a norm on a vector space, and all four togther constitute the definition of a norm on an algebra.)

(1) If $a \in M_n$, then $\|a\|$ is a nonnegative real number, and is zero if and only if $a = 0$.
(2) For $a, b \in M_n$, we have $\|a + b\| \leq \|a\| + \|b\|$.

Date: 14 November 2012.

Research partially supported by NSF grants DMS-9706850 and DMS-0701076.

(3) For $a \in M_n$ and $\lambda \in \mathbb{C}$, we have $\|\lambda a\| = |\lambda| \cdot \|a\|$.
(4) For $a, b \in M_n$, we have $\|ab\| \leq \|a\| \cdot \|b\|$.

It is important to realize that, in general, we do *not* have equality in (4).

We follow the convention, usual in operator algebras, that the identity matrix (of any size) is denoted by 1. (If it is necessary to specify the size, we use 1_n for the $n \times n$ identity matrix. Moreover, if $\lambda \in \mathbb{C}$, then $\lambda \cdot 1$ is abbreviated to λ. Thus, the expression which would typically be written $\lambda I - a$ in a linear algebra course becomes $\lambda - a$.

Warning: Large parts of this file have not been proofread at all, and some references are incomplete or missing entirely. Some additional information has been added in the last section, and not yet properly incorporated into the main text.

1. Approximate unitary equivalence

Roughly speaking, the question is as follows. This question was raised about 10 years ago, and I don't know for sure that it has not been answered since.

Question 1.1. Let u and v be randomly chosen (independent and uniform with respect to Haar measure) unitary matrices of the same large size n. How close are they likely to be to being unitarily equivalent? That is, find a bound δ_n such that, with high probability, there is a unitary z such that $\|zuz^* - v\| < \delta_n$.

The number $\inf_{z \in U(M_n)} \|zuz^* - v\|$ has a simple description in terms of the eigenvalues of u and v, based on the fact that both are unitarily diagonalizable. Specifically, let $\lambda_1, \lambda_2, \ldots, \lambda_n$ be the eigenvalues of u, repeated according to multiplicity, and let $\mu_1, \mu_2, \ldots, \mu_n$ be the eigenvalues of v, repeated according to multiplicity. Then there are unitaries x and y such that

$$xux^* = \operatorname{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n) \quad \text{and} \quad yvy^* = \operatorname{diag}(\mu_1, \mu_2, \ldots, \mu_n)$$

(diagonal matrices with the specified entries on the diagonal). One easily checks that

$$\|(y^*x)u(y^*x)^* - v\| = \|\operatorname{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n) - \operatorname{diag}(\mu_1, \mu_2, \ldots, \mu_n)\| = \max_{1 \leq k \leq n} |\lambda_k - \mu_k|.$$

So one should choose the listing so as to minimize the maximum of the errors on the right. Given a specific listing, this means applying a permutation to one of the lists. In fact, this gives the best possible result for $\|zuz^* - v\|$ for arbitrary unitaries z. In the following theorem, S_n denotes the symmetric group on n letters, that is, the set of bijections of $\{1, 2, \ldots, n\}$.

Part (1) of the following theorem is Theorem 13.6 of [1]. Something that approaches Part (2) is in a second proof of that theorem, in Section 17 of [1]. I believe it can be proved by a combinatorial argument using only Part (1), but I have not checked.

Theorem 1.2. Let $n \in \mathbb{Z}_{>0}$, and let $u, v \in M_n$ be unitaries. Let

$$\lambda_1, \lambda_2, \ldots, \lambda_n \quad \text{and} \quad \mu_1, \mu_2, \ldots, \mu_n$$

be the eigenvalues of u and v, repeated according to multiplicity.

(1) We have

$$\inf_{z \in U(M_n)} \|zuz^* - v\| = \min_{\sigma \in S_n} \max_{1 \leq k \leq n} |\lambda_k - \mu_{\sigma(k)}|.$$

(2) If the eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$ are listed in cyclic order around the unit circle, then the permutation σ which gives the minimum in (1) has the property that $\mu_{\sigma(1)}, \mu_{\sigma(2)}, \ldots, \mu_{\sigma(n)}$ are listed in cyclic order around the unit circle.

The problem with Part (2) is that there is no immediate way to determine the starting point of the cyclic order to use for $\mu_{\sigma(1)}, \mu_{\sigma(2)}, \ldots, \mu_{\sigma(n)}$.

It turns out that the eigenvalues of random unitaries tend to be much more uniformly distributed around the unit circle than independent randomly chosen complex numbers of absolute value 1. (See, for example, Section IV of [3], although this is not the original reference for the random unitary matrices.) Compare Figures 1 and 2. This means that the number δ_n in Question 1.1 is smaller than it would be if the distribution of eigenvalues were close to random.

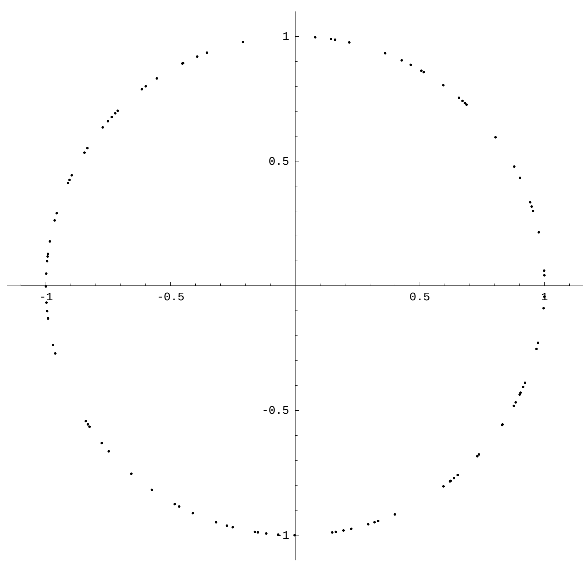

FIGURE 1. 100 independent uniform random points on the unit circle.

A numerical experiment related to Question 1.1 (or to other questions described here) requires a method of constructing uniformly distributed random unitaries. Fortunately, this is fairly easy. One starts by choosing $2n^2$ real numbers, say $\alpha_{j,k}$ and $\beta_{j,k}$ for $j, k = 1, 2, \ldots, n$, each independently chosen from the standard normal distribution. One uses these to construct n^2 elements of $\mathbb{C}^n$:

$$
\begin{aligned}
\xi_1 &= \big(\alpha_{1,1} + \beta_{1,1} i,\ \alpha_{2,1} + \beta_{2,1} i,\ \ldots, \alpha_{n,1} + \beta_{n,1} i\big), \\
\xi_2 &= \big(\alpha_{1,2} + \beta_{1,2} i,\ \alpha_{2,2} + \beta_{2,2} i,\ \ldots, \alpha_{n,2} + \beta_{n,2} i\big), \\
&\ldots, \\
\xi_n &= \big(\alpha_{1,n} + \beta_{1,n} i,\ \alpha_{2,n} + \beta_{2,n} i,\ \ldots, \alpha_{n,n} + \beta_{n,n} i\big).
\end{aligned}
$$

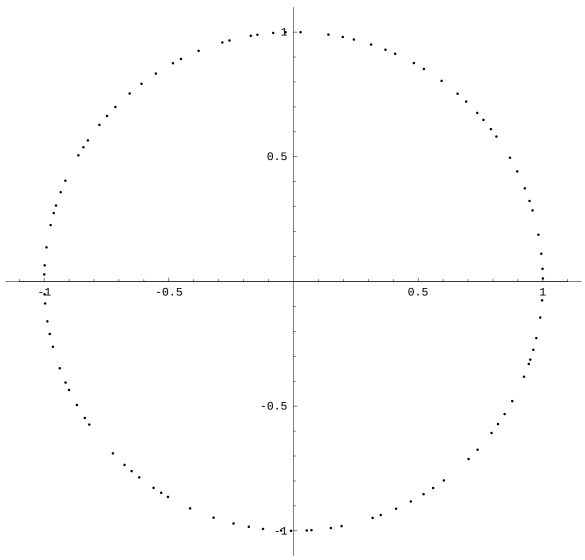

FIGURE 2. Eigenvalues of a random unitary of size 100.

Then one applies the Gram-Schmidt procedure to convert $\xi_1, \xi_2, \ldots, \xi_n$ to an orthonormal basis:

$$\begin{aligned}
\eta_1 &= \left(\frac{1}{\|\xi_1\|} \right) \xi_1, \\
\eta_2^{(0)} &= \xi_2 - \langle \xi_2, \eta_1 \rangle \eta_1, \\
\eta_2 &= \left(\frac{1}{\|\eta_2^{(0)}\|} \right) \eta_2^{(0)}, \\
\eta_3^{(0)} &= \xi_3 - \langle \xi_3, \eta_1 \rangle \eta_1 - \langle \xi_3, \eta_2 \rangle \eta_2, \\
\eta_3 &= \left(\frac{1}{\|\eta_3^{(0)}\|} \right) \eta_3^{(0)}, \\
&\ldots, \\
\eta_n^{(0)} &= \xi_n - \langle \xi_n, \eta_1 \rangle \eta_1 - \langle \xi_n, \eta_2 \rangle \eta_2 - \cdots - \langle \xi_n, \eta_{n-1} \rangle \eta_{n-1}, \\
\eta_n &= \left(\frac{1}{\|\eta_n^{(0)}\|} \right) \eta_n^{(0)}.
\end{aligned}$$

Now get the random unitary as the $n \times n$ matrix chose columns are $\eta_1, \eta_2, \ldots, \eta_n$. (It is just as good to take these to be the rows instead of the columns.)

In principle, it is possible that the elements $\xi_1, \xi_2, \ldots, \xi_n$ are linearly dependent, so that the Gram-Schmidt process can't be carried out. In theory, this happens with probability zero, so can be ignored.

It might be interesting to compare the asymptotics of approximate unitary equivalence of random unitaries with the asymptotics of the result of applying the matching of Theorem 1.2 to collections of randomly chosen points on the circle. (Again, this might turn out to be known.)

2. The asymptotics of the smallest gap

The problem discussed here was motivated by an approach to the problem discussed in Section 3 (the theoretical version of the idea in Remark 3.11), but it isn't at all clear that it is actually helpful.

Question 2.1. How does the expected value of the smallest gap between eigenvalues of a random $n \times n$ unitary matrix behave as $n \to \infty$?

To discuss this, it is convenient to have some notation.

Definition 2.2. Let $n \in \mathbb{Z}_{>0}$, and let $u \in M_n$ be unitary. Define the *smallest eigenvalue gap* $\gamma(u)$ of u to be

$$\gamma(u) = \min \big(\big\{ |\lambda_1 - \lambda_2| \colon \text{there are linearly independent } \xi_1, \xi_2 \in \mathbb{C}^n \\ \text{such that } u\xi_1 = \lambda_1 \xi_1 \text{ and } u\xi_2 = \lambda_2 \xi_2 \big\}.$$

If all the eigenvalues of u have multiplicity one, then $\gamma(u)$ is the smallest difference of distinct eigenvalues. If u has an eigenvalue of multiplicity more than one, then $\gamma(u) = 0$. In particular, the minimum in the definition of $\gamma(u)$ is over a finite set.

Definition 2.3. For $n \in \mathbb{Z}_{>0}$, define the *expected smallest eigenvalue gap* γ_n to be the expected value of $\gamma(u_n)$ as u varies over the $n \times n$ unitary matrices, with the uniform distribution.

We want to know about the behavior of γ_n as $n \to \infty$, and related questions. The obvious thing to observe is that an $n \times n$ unitary u has n eigenvalues, and the largest possible value of $\gamma(u)$ is obtained if they are equally spaced around the circle. Therefore $\gamma_n < 2\pi/n$. How much less than $2\pi/n$ is it?

A good answer would be to find a number $r > 0$ for which there is a constant $C > 0$ such that γ_n behaves like $C n^{-r}$ as $n \to \infty$. That is,

$$C = \lim_{n \to \infty} \frac{\gamma_n}{n^{-r}}$$

exists. This may well be too much to hope for; perhaps the best one can do is find $r > 0$ such that for every $\varepsilon > 0$ we have

$$\lim_{n \to \infty} \frac{\gamma_n}{n^{-r-\varepsilon}} = \infty \quad \text{and} \quad \lim_{n \to \infty} \frac{\gamma_n}{n^{-r+\varepsilon}} = 0.$$

One might also ask for something slightly different: to find $r > 0$ for which there is a constant $C > 0$ such that, as n gets large, for a random $n \times n$ unitary matrix u the probability that $\gamma(u) < C n^{-r}$ goes to zero.

According to my understanding of discussions with Chris Sinclair, problems of this nature fall in the area known as "order statistics", and in principle there are methods for obtaining an answer from the (known) joint distribution of the eigenvalues. It is possible that the answer is even published, although I have not seen anything like this.

What I am suggesting here is to try to learn something about the behavior of γ_n through numerical experiments. One could, for example, for small n estimate numerically the number γ_n by choosing many random $n \times n$ unitary matrices u and finding $\gamma(u)$ for each of them. Or one could similarly estimate the number ρ_n such that the probability that a random $n \times n$ unitary matrix u has $\gamma(u) < \rho_n$ is $\frac{1}{2}$. Probably γ_n and ρ_n behave very similarly, but this is not guaranteed.

It would be interesting to compare γ_n (and ρ_n) with the analogous quantities obtained using sets of n independent randomly chosen points on the circle instead of the eigenvalues of randomly chosen $n \times n$ unitary matrices. (Presumably methods of order statistics could give some theoretical results here.)

3. Approximately commuting unitary matrices

The following theorem is due to Huaxin Lin [9]. A much easier proof was provided by Friis and Rørdam [7], and some explicit estimates and methods for actually finding the nearby commuting matrices can be found in [8].

Theorem 3.1. For every $\varepsilon > 0$ there is $\delta > 0$ such that the following holds. Let n be a positive integer, and let $a_1, a_2 \in M_n$ be selfadjoint and satisfy

$$\|a_1\| \leq 1, \quad \|a_2\| \leq 1, \quad \text{and} \quad \|a_1 a_2 - a_2 a_1\| < \delta.$$

Then there exist selfadjoint $b_1, b_2 \in M_n$ such that

$$\|b_1\| \leq 1, \quad \|b_2\| \leq 1, \quad \|a_1 - b_1\| < \varepsilon, \quad \|a_2 - b_2\| < \varepsilon, \quad \text{and} \quad b_1 b_2 = b_2 b_1.$$

The restriction $\|a_1\|, \|a_2\| \leq 1$ is necessary because the norm of the commutator scales differently than the norms of the elements. The most important point here is that δ depends only on ε, not on n. This theorem solved a long standing open problem.

If one replaces selfadjoint matrices with unitaries, the analogous result is false. The standard counterexample is provided by the well known Voiculescu matrices, and was first found in [13]. Let $n > 0$, and let $\zeta_n = \exp(2\pi i/n)$, a primitive n-th root of 1. Then the Voiculescu matrices are given by

$$w_n = \begin{pmatrix} 1 & 0 & 0 & \cdots & \cdots & 0 \\ 0 & \zeta_n & 0 & \cdots & \cdots & 0 \\ 0 & 0 & \zeta_n^2 & \cdots & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & & \vdots \\ \vdots & \vdots & \vdots & & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & \cdots & \zeta_n^{n-1} \end{pmatrix} \quad \text{and} \quad s_n = \begin{pmatrix} 0 & 0 & \cdots & \cdots & 0 & 1 \\ 1 & 0 & \cdots & \cdots & 0 & 0 \\ 0 & 1 & \cdots & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & & \vdots & \vdots \\ \vdots & \vdots & & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & \cdots & 1 & 0 \end{pmatrix}.$$

That is, w_n is a diagonal matrix with the powers of ζ_n down the diagonal, and s_n is a cyclic shift. It is easy to check that

$$\|s_n v_n - v_n s_n\| = \|s_n v_n s_n^* - v_n\| = |\zeta_n - 1| < 2\pi/n.$$

It can be shown that there do not exist unitaries x close to v and y close to s such that $xy = yx$. Some of the theory behind this is discussed below, but we point out here an elementary proof (using nothing beyond winding numbers) in [5] which shows that if $x, y \in M_n$ are any matrices (not necessarily unitary, or even normal) with $xy = yx$, and if $n \geq 7$, then

$$\max\left(\|x - v_n\|, \|y - s_n\|\right) \geq \sqrt{2 - |\zeta_n - 1|} - 1.$$

The main point of this section is to discuss the following question and possible numerical experiments to test it. The question will be formulated more precisely below.

Question 3.2. Is it true that "most" pairs of approximately commuting large unitary matrices are close to pairs of commuting unitary matrices?

The answer to this question is not known.

Given two approximately commuting unitaries, one can test whether they are close to commuting unitaries with the help of some invariants. See [6] for a discussion of several of these, and [4] for one improvement. The most easily computable one (both theoretically and numerically) is the winding number invariant, defined as follows.

Definition 3.3. Let $n > 0$, and let $u, v \in U(M_n)$ satisfy $\|uv - vu\| < 2$. We define the *winding number invariant* $\omega(u, v)$ to be the winding number about 0 of the closed path in $\mathbb{C} \setminus \{0\}$ given by $\alpha \mapsto \det\big((1-\alpha)uv + \alpha vu\big)$ for $\alpha \in [0, 1]$.

The condition $\|uv - vu\| < 2$ ensures that $\big\|uv - [(1-\alpha)uv + \alpha vu]\big\| < 1$ for $\alpha \in \left[0, \frac{1}{2}\right]$ and $\big\|vu - [(1-\alpha)uv + \alpha vu]\big\| < 1$ for $\alpha \in \left[\frac{1}{2}, 1\right]$. This ensures that $(1-\alpha)uv + \alpha vu$ is always invertible, so that its determinant really is never zero.

There are also various forms of a K-theoretic invariant that we call $\kappa(u, v)$. Several equivalent versions are given in [6] for $\|uv - vu\|$ small, and an extension (less elementary) valid for all u and v with $\|uv - vu\| < 2$ is given in [4]. It is shown in [6] and in Theorem 3.2 of [4] that these invariants agree with the winding number invariant above.

The winding number invariant is the only obstruction in an asymptotic sense.

Theorem 3.4 (Corollary 6.15 of [2])**.** For every $\varepsilon > 0$ there is $\delta > 0$ such that the following holds. Let n be a positive integer, and let $u_1, u_2 \in U(M_n)$ satisfy

$$\|u_1u_2 - u_2u_1\| < \delta \quad \text{and} \quad \omega(u_1, u_2) = 0.$$

Then there exist $v_1, v_2 \in U(M_n)$ such that

$$\|u_1 - v_1\| < \varepsilon, \quad \|u_2 - v_2\| < \varepsilon, \quad \text{and} \quad v_1v_2 = v_2v_1.$$

Unfortunately, we do not know anything about how δ depends on ε. The following is a presumably difficult theoretical problem.

Problem 3.5. In Theorem 3.4, find an explicit formula for a possible value of δ in terms of ε.

Nevertheless, it seems reasonable to hope that the winding number invariant gives the right answer when the norm of the commutator is not too large and the matrix size is large. We therefore propose the following numerical experiments:

Problem 3.6. Determine experimentally whether "most" pairs of approximately commuting large unitary matrices have winding number invariant equal to zero. More generally, experimentally estimate the statistical distribution of the values of the winding number invariant for pairs of large unitary matrices (u, v) such that $\|uv - vu\| < 2$.

Problem 3.7. Experimentally estimate the asymptotics of the quantities in Problem 3.6 as the matrix size becomes large. For example, for a carefully chosen scaling factor ε_n (possibly $\varepsilon_n = \varepsilon/\sqrt{n}$, but this is not clear), find the asymptotic distribution of the winding number invariant for pairs of $n \times n$ unitary matrices (u, v) such that $\|uv - vu\| < \varepsilon_n$.

Remark 3.8. The winding number invariant $\omega(u, v)$ of Definition 3.3 is in principle numerically computable, although it is not clear how easily it can be done in practice. For $\alpha \in [0, 1]$ set $\gamma(\alpha) = \det\big((1-\alpha)uv + \alpha vu\big)$. The formula is

$$\omega(u, v) = \frac{1}{2\pi i}\int_{\gamma}\frac{1}{\zeta}\, d\zeta = \frac{1}{2\pi i}\int_0^1 \frac{\gamma'(\alpha)}{\gamma(\alpha)}\, d\alpha.$$

Determinants are computed by row reduction. I don't know whether it is better to numerically estimate the derivative, or to derive an algebraic form for it in terms of other determinants. There are a number of numerical methods for calculating integrals. One need not compute the integral very accurately, since it is known ahead of time that the answer must be an integer.

It seems very hard to generate randomly pairs of approximately commuting unitaries. One can easily generate random unitaries. To get a random $n \times n$ unitary matrix, one simply chooses $2n^2$ real numbers independently from the standard normal distribution, uses them as the real and imaginary parts of the entries of n vectors in $\mathbb{C}^n$, and applies the Gram-Schmidt process to turn these vectors into an orthonormal basis.

However, most pairs will not approximately commute, and the proportion that do will decrease exponentially with the dimension.

Here are several ideas that, by themselves, don't solve this difficulty, but which might help. The first two came from discussion with Benjamin Morris in Jan. 2005.

Remark 3.9. The "Metropolis Algorithm" is essentially a very general method for generating random elements of connected sets. The element depends on some parameters (here, the $2n^2$ entries of two $n \times n$ unitary matrices). Start with an element of the set (here, $\{(u, v) \in U(M_n) \colon \|uv - vu\| < \varepsilon\}$). Change one of the parameters, chosen at random, by a random small amount, and push the perturbed matrix back into the unitary group (using functional calculus). If the result is still in the set, replace the old element by the new one; otherwise, keep the old element. In a connected set, if this procedure is run long enough, the distribution will be uniform. In a very long and thin set, which is quite possibly what we have, the time required to approach uniformity will be very large. One with "bottlenecks" may be even worse. Also, the set of pairs of almost commuting unitaries is not connected; pairs for which the winding number invariant is different must be in different connected components. It is not clear whether the set of pairs of almost commuting unitaries with fixed winding number invariant is connected.

Remark 3.10. There is a kind of "doubling" method for estimating volumes. We want to estimate the volume of a set S_0, known to be contained in some space with finite volume. We choose a larger set S_1 whose volume should be very roughly twice that of S_0. Use a Monte Carlo method to estimate $\mathrm{vol}(S_0)/\mathrm{vol}(S_1)$ as the conditional probability that a point is in S_0 given that it is in S_1. Then choose a larger set S_2, etc. Probably one will choose S_{n+1} to be a suitable neighborhood of S_n, or else choose the S_n to be suitable neighborhoods of S_0.

Remark 3.11. The following refinement should at least substantially speed up the brute force approach. Choose a unitary u_0 at random, and diagonalize it, in such a way that the diagonal of the resulting unitary u is the eigenvalues of u_0 listed in cyclic order around the unit circle. We get $u = zu_0z^*$ for some unitary z. Now choose v at random. The probability that $\|uv-vu\| < \varepsilon$, and the probability that u and v are close to commuting unitaries, should be the same as if both were chosen completely at random. The reason is that if v is chosen uniformly at random, then z^*vz is also random and uniformly distributed.

For a very large proportion of the choices of u, the diagonal entries will be fairly close to being evenly spaced around the circle. Now some easy tests will show that many choices of v do not approximately commute with u. For example, if any entry of v a significant distance from the diagonal is fairly large, it probably follows that $\|uv - vu\|$ can't be very small. (Lemma 4.3 of [12] shows that two almost commuting unitary matrices are simultaneously approximately diagonalizable, but this probably doesn't help much, since it isn't clear how the basis is related to the given one. The main result of [11] gives some information about what happens when the diagonal entries are not evenly spaced.)

Presumably one can get reasonable numerical estimates of probabilities by using many choices of u, and for each fixed choice of u, using many choices of v. In a different direction, it might be possible to numerically generate eigenvalue lists for random unitaries directly, much more quickly than diagonalizing randomly chosen unitaries.

4. Additional information

This section contains some additional information which has not yet properly incorporated into the main text.

Marius Dadarlat's talk slides from the June 2011 CRM conference have various versions or generalizations of Loring and Exel-Loring formulas for the K-theoretic obstruction to perturbing almost commuting unitaries to commuting ones.

The following is from an email message from Chris Sinclair of 21 June 2011.

My question: Second, do you know of any way of producing a large number of lists of eigenvalues chosen from this distribution that is more efficient than generating a large number of random unitaries and diagonalizing them? This is wanted for numerical experiments. It is easy to generate a random unitary in M_n: Choose n elements of $\mathbb{C}^n$ whose real and imaginary parts are independent normally distributed with the same standard deviation and with mean zero, and apply Gram-Schmidt to orthonormalize them. But then there is numerical work to find eigenvalues.

Response: The distribution of eigenvalues is known, so there must be some standard sampling method which would be applicable. However, I am not an expert on this sort of statistics, so I really can't tell you. Another possible approach is the following: Consider the set of polynomials with all roots on the unit circle. One can parametrize this set in a natural way so that the coefficient vectors of such polynomials can be identified with a subset of $\mathbb{C}^M$ (for some appropriate value of M). The roots of a polynomial chosen uniformly in this set will have the same distribution as the eigenvalues of a random unitary matrix. (I hope I am remembering this correctly. You might look at http://arxiv.org/pdf/math/0511397). One could then do some sort of selection/rejection regime to select such a polynomial uniformly. There are bounds for the coefficients, which makes the search space bounded. See:

Sinclair, C. D., & Vaaler, J. D. (2008). Self-Inversive Polynomials with All Zeros on the Unit Circle. In J. McKee & C. Smyth (Eds.), Number Theory and Polynomials: Proceedings of the workshop held at Bristol University, Bristol, April 3–7, 2006, London Mathematical Society Lecture Note Series (Vol. 352, pp. xiv+349). Cambridge: Cambridge University Press. doi:10.1017/CBO9780511721274

I'm not sure this method will be useful however, since the volume of the set of such polynomials is rather small, especially when the degree is large.

My question: Third, do you know a citable reference for generating random unitaries numerically?

Response: No.

My question: Fourth, have you ever heard of any work on the following question: Given two $n \times n$ random unitaries u and v, what are the asymptotics of how close they are to being unitarily equivalent? That is, consider, say, the expected value of the minimum of $\|u - zvz*\|$ over all unitaries z, and ask how this quantity behaves as $n \to \infty$. (It is known that the minimum of $\|u - zvz*\|$ depends only on the eigenvalues of u and v.)

Response: Huaxin asked me a question along these lines once. However, I haven't had the time to really consider it. I know of no results in this direction, though I'm not really an expert in circular ensembles.

My question: As a possible contrast with some of this, is anything known about the asymptotics of the following: Choose n points on the circle, independently and uniformly according to arc length measure. What happens to the expected value of the minimum distance between two of these point, as $n \to \infty$?

Response: This must be known, or at least knowable. This follows under the guise of order statistics. Once one knows the distribution of the gaps between points, then one can "easily" get the distribution of the smallest gap and therefore the expectation. This would be a good exercise in order statistics. Determining the asymptotics of this quantity would then, in principle, be doable. For the Circular Unitary Ensembles (random unitaries) the asymtptotic distribution of gaps (scaled so that the expected gap size is, say, 1) is known. I suspect the asymptotics of the smallest gap is known too, but I don't know of a reference off the top of my head. By universality, the answer must be the same as for the Gaussian Unitary Ensemble. You might look in Mehta's book Random Matrices, which has a wealth of such facts (though his proofs are often not rigorous).

References

[1] R. Bhatia, *Perturbation Bounds for Matrix Eigenvalues*, Pitman Research Notes in Math. no. 162, Longman Scientific and Technical, Harlow, Britain, 1987.

[2] S. Eilers, T. A. Loring, and G. K. Pedersen, *Morphisms of extensions of C*-algebras: Pushing forward the Busby invariant*, Adv. Math. **147**(1999), 74–109.

[3] F. J. Dyson, *Statistical theory of the energy levels of complex systems. I*, J. Mathematical Phys. **3**(1962), 140–156.

[4] R. Exel, *The soft torus and applications to almost commuting matrices*, Pacific J. Math. **160**(1993), 207–217.

[5] R. Exel and T. A. Loring, *Almost commuting unitary matrices*, Proc. Amer. Math. Soc. **106**(1989), 913–915.

[6] R. Exel and T. A. Loring, *Invariants of almost commuting unitaries*, J. Funct. Anal. **95**(1991), 364–376.

[7] P. Friis and M. Rørdam, *Almost commuting self-adjoint matrices – a short proof of Huaxin Lin's theorem*, J. reine angew. Math. **479**(1996), 121–132.

[8] M. B. Hastings, *Making almost commuting matrices commute*, Commun. Math. Phys. **291**(2009), 321–345.

[9] H. Lin, *Almost commuting selfadjoint matrices and applications*, pages 193–233 in: *Operator Algebras and their Applications (Waterloo, ON, 1994/1995)*, Fields Inst. Commun. vol. 13, Amer. Math. Soc., Providence, RI, 1997.

[10] M. L. Mehta, *Random matrices* (3rd ed.), Pure and Applied Mathematics (Amsterdam), vol. 142, Elsevier/Academic Press, Amsterdam, 2004.

[11] T. J. Osborne, *Almost commuting unitaries with spectral gap are near commuting unitaries*, Proc. Amer. Math. Soc. **137**(2009), 4043–4048.

[12] M. Steiner, *Variations on a theme of Frobenius about almost commuting unitary matrices*, Linear Algebra Appl. **400**(2005), 1–13.

[13] D. Voiculescu, *Asymptotically commuting finite rank unitary operators without commuting approximants*, Acta Sci. Math. (Szeged) **45**(1983), 429–431.

Department of Mathematics, University of Oregon, Eugene OR 97403-1222, USA.